

# Radioactive Decay and Half-Life

Answer key with worked solutions

## A. Understanding

### 1. Definitions.

(a) The spontaneous decay of **unstable nuclei**, which emit radiation (alpha, beta, positron or gamma). (b) The **time for half** of the radioactive nuclei in a sample (and its activity) to decay. (c) The decay of any one nucleus is **random**, but in a very large number of nuclei a constant **fraction** decays in each unit of time, so the sample as a whole follows the half-life.

## B. Nuclei and nuclear equations

### 2. Protons and neutrons.

(a) **92 p, 143 n** (b) **27 p, 33 n** (c) **53 p, 78 n** (d) **86 p, 136 n**. Neutrons =  $A - Z$ .

### 3. Completed equations.

(a)  ${}^{222}_{86}\text{Rn}$  (b)  ${}^{131}_{54}\text{Xe}$  (c)  ${}^{18}_8\text{O}$  (d)  ${}^{60}_{28}\text{Ni}$  (e)  ${}^{235}_{92}\text{U}$  (f)  ${}^{40}_{20}\text{Ca}$ . In each, the mass numbers and atomic numbers balance; for example (e):  $239 = 235 + 4$  and  $94 = 92 + 2$ .

### 4. Identify X.

(a)  $A = 0$ ,  $Z = -1$ : a **beta particle**,  ${}^0_{-1}\text{e}$ . (b)  $A = 4$ ,  $Z = 2$ : an **alpha particle**,  ${}^4_2\text{He}$ . (c)  $A = 0$ ,  $Z = +1$ : a **positron**,  ${}^0_{+1}\text{e}$ .

### 5. Uranium-238 series.

${}^{238}_{92}\text{U} \longrightarrow {}^{234}_{90}\text{Th} + {}^4_2\text{He}$ ;  ${}^{234}_{90}\text{Th} \longrightarrow {}^{234}_{91}\text{Pa} + {}^0_{-1}\text{e}$ ;  ${}^{234}_{91}\text{Pa} \longrightarrow {}^{234}_{92}\text{U} + {}^0_{-1}\text{e}$ . Final nucleus: **uranium-234**,  ${}^{234}_{92}\text{U}$  (an isotope of the starting element).

## C. Types of radiation

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### 6. Comparing radiation.

**Alpha:** a helium nucleus, charge +2; stopped by paper or skin. **Beta:** an electron, charge -1; stopped by a few mm of aluminium. **Gamma:** electromagnetic radiation, no charge; only reduced by thick lead or concrete. Inside the body, an alpha emitter deposits all its energy in a very short distance in living cells, and alpha particles ionize strongly, so they cause the most damage there.

## D. Half-life

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### 7. Technetium-99m.

$$n = \frac{24.04 \text{ h}}{6.01 \text{ h}} = 4.00 \text{ half-lives}$$
$$\text{activity} = 800 \text{ MBq} \times \left(\frac{1}{2}\right)^{4.00} = \frac{800 \text{ MBq}}{16} = \mathbf{50.0 \text{ MBq}}$$

### 8. Half-life from data, and radon.

(a)  $1600 \text{ Bq} \rightarrow 800 \text{ Bq} \rightarrow 400 \text{ Bq}$  is **2 half-lives**, so  $t_{1/2} = \frac{30.0 \text{ min}}{2} = \mathbf{15.0 \text{ min}}$

(b)  $n = \frac{10.0 \text{ d}}{3.82 \text{ d}} = 2.618$ ; fraction remaining  $= \left(\frac{1}{2}\right)^{2.618} = 0.163$ , so **16.3 %** remains.

## E. Challenge

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### 9. Radiocarbon age.

$$\frac{N_0}{N} = \frac{0.230 \text{ Bq/g}}{0.0850 \text{ Bq/g}} = 2.706 \text{ (the units cancel)}$$
$$n = \frac{\ln 2.706}{\ln 2} = \frac{0.9955}{0.6931} = 1.436 \text{ half-lives}$$

$t = 1.436 \times 5730 \text{ yr} = \mathbf{8.23 \times 10^3 \text{ yr}}$  (about 8230 years). Check: the activity has fallen to 37.0 %, between 50 % and 25 %, so the age lies between 1 and 2 half-lives. ✓